EXPERIMENT 2

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USER INTERFACE DESIGN

MEMORY RECALL TASK – UI DESIGN AND CHUNKING

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1. Introduction

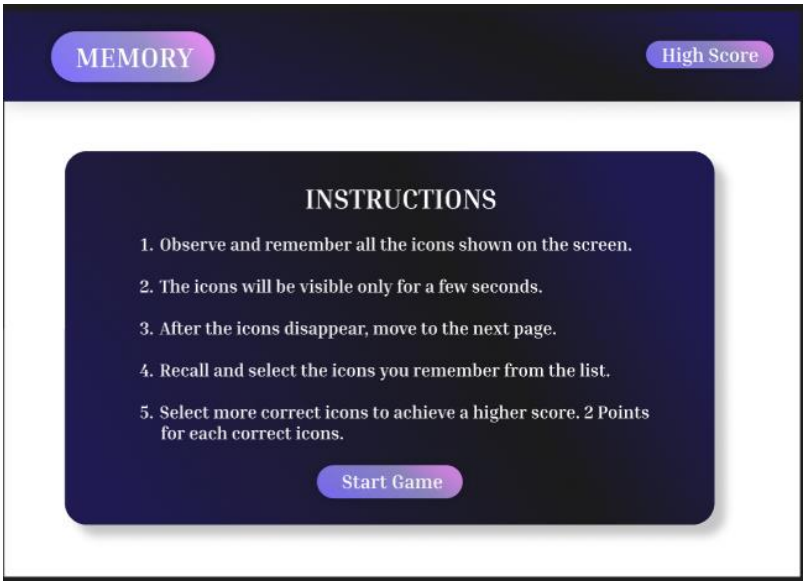
This project focuses on the design and analysis of a **Memory Recall Task UI**, where users are required to observe visual elements such as **icons or text chunks**, memorize them within a limited time, and later recall the same information. The core objective of this design is to **evaluate the effect of chunking on user memory**. Chunking is a well-known cognitive strategy that improves memory retention by grouping information into smaller, meaningful units.

The user interface is divided into four main frames:

1. Instruction Page
2. Chunking Phase
3. Recall Phase
4. Result Page

Each frame is carefully designed to support memory encoding, recall accuracy, and user engagement while minimizing cognitive overload.

FRAME 1: INSTRUCTION PAGE



Chunking Analysis of the Instruction Page

The Instruction Page introduces the task to the user and prepares them mentally for the memory challenge. This screen effectively applies the **chunking principle** by breaking down information into structured, easy-to-understand units.

Key Design and Cognitive Features

- The instructions are divided into **six clearly numbered steps**, which helps users process information sequentially.
- Each step communicates **only one key action**, preventing information overload.
- The instructions are logically grouped into phases:
  - **Observation Phase (Steps 1–2):** Explains what the user will see.
  - **Memorization Strategy (Step 3):** Encourages focus and attention.
  - **Recall Phase (Steps 4–5):** Explains how recall will be tested.
  - **Performance Metric (Step 6):** Emphasizes accuracy over speed.

Visual Hierarchy and UI Design

- A bold and prominent title clearly identifies the task.
- Adequate spacing and structured text improve readability.
- The **"Start" button** is visually highlighted, guiding the user toward the next action.
- The use of a **bee-themed visual element** adds a playful touch without distracting from the task.

Cognitive Benefits

- Short, direct sentences improve comprehension.
- Clear time constraints prepare users mentally for the challenge.
- The structured layout reduces cognitive load and improves user confidence before starting the task.

FRAME 2: CHUNKING PHASE

Analysis of the Memory Encoding Screen

The Chunking Phase is the **memory encoding stage**, where users observe and mentally group visual elements before recall.

Purpose of the Screen

- This phase allows users to **encode visual information** into short-term memory.
- The design encourages users to apply chunking strategies by identifying patterns and similarities among items.

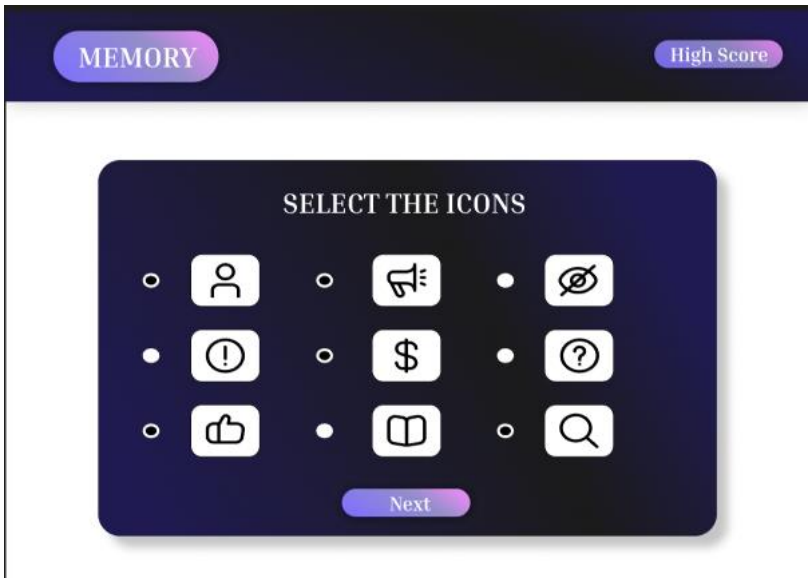
Key UI Components

- **Countdown Timer:**
  - Displays the limited observation time.
  - Reinforces urgency and focus.
- **Progress Indicator:**
  - Visually represents remaining time.
  - Helps users manage their attention efficiently.
- **Grid of Icons / Words:**
  - Items are displayed in a structured grid.
  - Icons are visually distinct yet suitable for grouping.
  - Encourages chunking based on:
    - Categories (e.g., animals, food, transport)
    - Repetition
    - Visual similarity
- **Mascot Element:**
  - A bee character adds engagement and thematic consistency.
  - Maintains user interest without affecting task performance.

Cognitive and UX Benefits

- Supports memory formation by grouping related items.
- Reduces mental effort by allowing users to remember clusters instead of individual items.
- Time-restricted viewing forces efficient memory strategies.

FRAME 3: RECALL PHASE



Analysis of the Selection Screen

The Recall Phase tests the user's ability to retrieve information stored during the Chunking Phase.

Purpose of the Screen

- This stage evaluates **short-term memory retention**.
- It measures the effectiveness of chunking strategies used by the user.

UI Components and Interaction Design

- A clear title and instruction guide users on what to do.
- A set of selectable options is displayed, including:
    - Correct items from the previous phase.
    - Distractor items to test false memory.
- Each option includes a **radio button**, allowing clear selection.
- The layout is balanced and uncluttered, supporting focus.

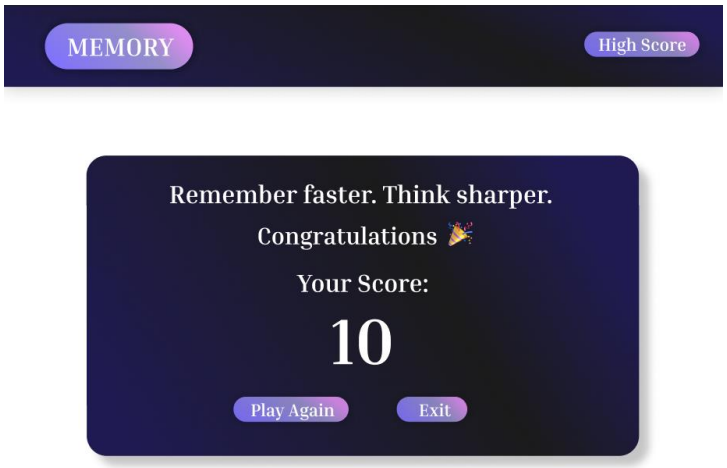
Recall Process

- Users analyze the available options.
- They select only the items they remember seeing earlier.
- The presence of distractors tests memory accuracy rather than guesswork.
- The **Submit button** finalizes the user's response.

Cognitive and UX Benefits

- Encourages deliberate recall instead of random selection.
- Tests the user's ability to distinguish correct memories from false ones.
- Clear selection mechanics improve usability and reduce errors.

FRAME 4: RESULT PAGE



Analysis of the Score and Feedback Screen

The Result Page provides feedback on the user's memory performance and allows them to decide the next action.

Purpose of the Screen

- Displays recall accuracy in a clear and understandable manner.
- Motivates users to improve performance through feedback.

Key UI Elements

- A prominent title reinforces task completion.
- The score is displayed clearly, showing correct answers out of total attempts.
- Action buttons include:
    - **Continue:** Proceed to the next level.
    - **Restart:** Retry the task for better performance.
    - **Exit:** End the session.
- The bee mascot continues to provide visual continuity and engagement.

Cognitive and UX Benefits

- Immediate feedback helps users evaluate their memory skills.
- Multiple action choices give users control over their learning process.
- Gamified elements reduce stress and enhance motivation.

Conclusion

The Memory Recall Task UI successfully demonstrates the application of **chunking as a cognitive strategy** to improve memory performance. Each screen is purposefully designed to support attention, memory encoding, recall accuracy, and user engagement. The structured flow from instruction to result ensures clarity and consistency, while visual and interactive elements maintain user interest without increasing cognitive load.

Overall, the design effectively balances **usability, cognitive psychology principles, and visual clarity**, making it a strong example of a memory-focused user interface.

Prototype Link:

https://www.figma.com/proto/3pljDmtYvGvBFg9iCz5Ewy/Memory?page-id=0%3A1&node-id=2-99&p=f&viewport=-1673%2C-1%2C0.2&t=A60a7fLCq1YNyE6j-1&scaling=min-zoom&content-scaling=fixed&starting-point-node-id=2%3A99&show-proto-sidebar=1